

Chapter 4 Homework: Using Script Files and Managing Data

Instructions:

- Write a script file for the exercises below.
- Make sure to write comments in your code and separate output display for clarity.
- Submit your script file in the following format: FirstName_LastInitial_ch4hw.m
Ex: Ryan Cao would submit his lab as: **ryan_c_ch4hw.m**

1.

Write a MATLAB program in a script file that calculate the average, standard deviation, and median of a list of grades as well as the number of grades on the list. The program asks the user (input command) to enter the grades as elements of a vector. The program then calculates the required quantities using MATLAB's built-in functions `length`, `mean`, `std`, and `median`.

The results are displayed in the Command Window in the following format:

"There are XX grades." where XX is the numerical value.

"The average grade is XX." where XX is the numerical value.

"The standard deviation is XX." where XX is the numerical value.

"The median grade is XX." where XX is the numerical value.

Execute the program and enter the following grades: 92, 74, 53, 61, 100, 42, 80, 66, 71, 78, 91, 85, 79, and 68.

2.

The balance of a loan, B , after n monthly payments is given by

$$B = A\left(1 + \frac{r}{1200}\right)^n - \frac{P}{r/1200}\left[\left(1 + \frac{r}{1200}\right)^n - 1\right]$$

where A is the loan amount, P is the amount of a monthly payment, and r is the yearly interest rate entered in % (e.g., 7.5% entered as 7.5). Consider a 5-year, \$20,000 car loan with 6.5% yearly interest that has a monthly payment of \$391.32. Calculate the balance of the loan after every 6 months (i.e., at $n = 6, 12, 18, 24, \dots, 54, 60$). Each time, calculate the percent of the loan that is already paid. Display the results in a three-column table, where the first column displays the month and the second and third columns display the corresponding value of B and percentage of the loan that is already paid, respectively.

Correction: Please display the percent of the loan remaining in the 3rd column.

3.

The intrinsic electrical conductivity σ of a semiconductor can be approximated by:

$$\sigma = e^{(C - \frac{E_g}{2kT})}$$

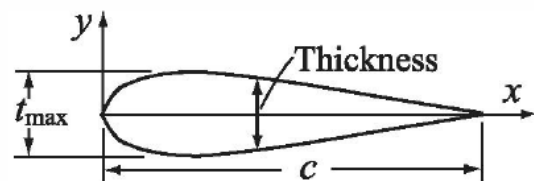
where σ is measured in $(\Omega \cdot m)^{-1}$, E_g is the band gap energy, k is Boltzmann's constant (8.62×10^{-5} eV/K), and T is temperature in kelvins. For Germanium, $C = 13.83$ and $E_g = 0.67$ eV. Write a program in a script file that calculates the intrinsic electrical conductivity for Germanium for various temperatures. The values of the temperature should be read from an xls spreadsheet using the `xlsread` command. The output should be presented as a table where the first column is the temperature and the second column is the intrinsic electrical conductivity.

Use the file: Germanium_data.xlsx

4.

The surface of many airfoils can be described with an equation of the form

$$y = \mp \frac{tc}{0.2} [a_0 \sqrt{x/c} + a_1(x/c) + a_2(x/c)^2 + a_3(x/c)^3 + a_4(x/c)^4]$$



where t is the maximum thickness as a fraction of the chord length c (e.g., $t_{\max} = ct$). Given that $c = 1$ m and $t = 0.2$ m, the following values for y have been measured for a particular airfoil:

x (m)	0.15	0.35	0.5	0.7	0.85
y (m)	0.08909	0.09914	0.08823	0.06107	0.03421

Determine the constants a_0 , a_1 , a_2 , a_3 , and a_4 . (Write a system of five equations and five unknowns, and use MATLAB to solve the equations.)

5.

The wind chill temperature, T_{wc} , is the air temperature felt on exposed skin due to wind. In U.S. customary units it is calculated by:

$$T_{wc} = 35.74 + 0.6215T - 35.75v^{0.16} + 0.4275Tv^{0.16}$$

where T is the temperature in degrees F, and v is the wind speed in mi/h. Write a MATLAB program in a script file that displays the following chart of wind chill temperature for given air temperature and wind speed in the Command Window:

		Temperature (F)								
		40	30	20	10	0	-10	-20	-30	-40
Speed (mi/h)										
10	34	21	9	-4	-16	-28	-41	-53	-66	
20	30	17	4	-9	-22	-35	-48	-61	-74	
30	28	15	1	-12	-26	-39	-53	-67	-80	
40	27	13	-1	-15	-29	-43	-57	-71	-84	
50	26	12	-3	-17	-31	-45	-60	-74	-88	
60	25	10	-4	-19	-33	-48	-62	-76	-91	

Output:

Problem 1:

Please enter the grades as a vector [x x x]: [92 74 53 61 100 42 80 66 71 78 91 85 79 68]

There are 14 grades.

The average grade is 74.3.

The standard deviation is 15.8.

The median grade is 76.0.

Problem 2:

	Balance	Remaining
Month	\$	%
6.00	18278.92	91.39
12.00	16501.14	82.51
18.00	14664.80	73.32
24.00	12767.96	63.84
30.00	10808.63	54.04
36.00	8784.76	43.92
42.00	6694.22	33.47
48.00	4534.80	22.67
54.00	2304.25	11.52
60.00	0.21	0.00

Problem 3:

Temperature	Intrinsic Conductivity
deg K	(ohm-m) ⁻¹
400	61.2
435	133.7
475	283.8
500	427.3
520	576.1
545	811.7

Problem 4:

The coefficients are:

a0=0.2969, a1=-0.1258, a2=-0.3526, a3=0.2861, a4=-0.1025

Problem 5:

	Temperature (F)								
	40	30	20	10	0	-10	-20	-30	-40
Speed (mi/hr)									
10	34	21	9	-4	-16	-28	-41	-53	-66
20	30	17	4	-9	-22	-35	-48	-61	-74
30	28	15	1	-12	-26	-39	-53	-67	-80
40	27	13	-1	-15	-29	-43	-57	-71	-84
50	26	12	-3	-17	-31	-45	-60	-74	-88
60	25	10	-4	-19	-33	-48	-62	-76	-91